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MILITARY SPECIFICATION

TEST SET, TACHOMETER DWELL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable tachometer-dwell test set, of solid state electronic design, with components mounted upon a printed circuit board. The test set measures revolutions per minute (rpm) and dwell on four, six, and eight cylinder automotive engines with 12 and 24 volt ignition systems. The set also measures point resistance. The complete test set is contained in a separate carrying case.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

STANDARDS

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

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3.2 Design. The test set shall be new and of the manufacturer's latest commercial design that shall meet the requirements specified herein. The test set shall be of solid state electronic component design. Components shall be mounted upon a printed circuit board. The test set shall measure and indicate on one multiscale meter the following:

(a) Engine speed in rpm ranging from 400 through 6,000 rpm on four, six, and eight cylinder engines.
(b) Ignition coil energizing buildup period in degrees of distributor camshaft lobe rotation, as electrical degrees of dwell on breaker point systems for four, six, and eight cylinder engines.
(c) Ignition point resistance on breaker point systems only.

The test set shall perform the above functions on the following types of ignition systems:

American Motors breakerless1/
Ford breakerless1/
Presto-Lite breakerless1/
Chrysler electronic1/
General Motors high energy ignition (HEI)1/
Presto-Lite breaker point
Military breaker point and ignition systems, including those with built-in ballast resistors
All standard commercial breaker point types

1/No dwell and no point resistance shall be required on systems without ignition points.

The test set shall be capable of performing satisfactorily on each of the ignition systems identified above, especially including either one of the M151 series of military vehicles (Truck, Utility, 1/4 Ton, M151A1, NSN 2320-00-763-1092 or M151A2, NSN 2320-00-177-9258) and one of the M880 series of military vehicles (Truck, Cargo, 1 1/4 Ton, 4 X 4, M880, NSN 2320-00-579-8942 or M890, 4 X 2, NSN 2320-00-579-8991). (See 6.4). The test set shall be electrically energized by only the 12 and 24 volt ignition current at the coil. The set shall automatically select the correct circuitry for the energizing current being made available to it at the moment of hookup. The test set shall be energized by only the primary, low voltage side of the ignition system current of the circuit being tested. No other internal or external power, or energizing source shall be permitted. Operation from the secondary, high voltage side of the ignition system is not permitted. Consistent and compatible with the type of ignition system being tested, each test set function (a, b, and c above) shall perform satisfactorily, in accordance with the requirements herein, on the systems listed above. The test set shall be housed in a sheet metal or acrylonitrile butadiene styrene (ABS) plastic case, or combination thereof. An additional metal or plastic carrying case with handle shall be furnished to carry and protect the test set during transit. A pigtail, with an electrical spade or bayonet type connection, shall be supplied with each test set to access the tachometer-distributor connection on the General Motors HEI system. An electrical hair-pin type clip shall be supplied with each test set for access to the Ford breakerless system.

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3.2.1 Standard breaker point type ignition systems. For purposes of demonstrating the test set upon standard breaker point type ignition systems, the following list comprises the least acceptable representative sample:

2 - Standard breaker point ignition systems, one each from two different automotive divisions of Ford Motor Corp., either four, six, or eight cylinder.

2 - Standard breaker point ignition systems, one each from two different divisions of General Motors Corp., either four, six, or eight cylinder.

1 - Standard breaker point ignition system from American Motors Corp., either four, six, or eight cylinder.

1 - Standard breaker point ignition system from Chrysler Corp., six cylinder.

In addition to the above, military breaker point ignition systems from the M151A1 or M151A2, and from the M880 or M890 series of military vehicles shall be included (see 3.2). The above sample shall total six ignition systems to represent standard breaker point systems only. The systems shall be representative of four, six, and eight cylinder engine ignition systems. (As an example, the manufacturer may select: a six lobe ignition system from Mercury and an eight lobe ignition system from Lincoln; a four lobe ignition system from Chevrolet and an eight lobe ignition system from Buick; a six lobe ignition system from Dodge; and a six lobe ignition system from American Motors.)

3.2.2 High energy ignition systems. The high energy ignition systems shall be represented by the following minimum requirements:

Four cylinder American Motors breakerless system.
Four, six, or eight cylinder Presto-Lite breakerless system.
Six cylinder Chrysler electronic system.
Eight cylinder Ford breakerless system.
Eight cylinder General Motors HEI system.
Four cylinder M151A1 or M151A2 military vehicles with electronic conversion kit installed.

A pigtail with an electrical spade/bayonet type connection shall be supplied with each test set to access the tachometer - distributor connection on the General Motors HEI system. An electrical hair pin clip shall be provided with each test set to access the Ford breakerless ignition system.

3.2.3 Presto-Lite breaker point system. A four, six, or eight cylinder Presto-Lite breaker point system shall be used to represent the Presto-Lite breaker point systems.

3.2.4 Incorrect connection and overload protection. The test set shall be designed to prevent damage to itself and any ignition system in the event the set is hooked up to the ignition system incorrectly, including reverse polarity. The test set as a whole, and each of its individual functions, shall be capable of withstanding electrical power input connected to the test leads when subjected to 120 volts alternating current circuit at 30 amps capacity and 24 volts direct current from a fully charged 500 amp hour battery for periods of from 5 to 15 seconds. The test set shall not incur damage when connected to any part of 12 or

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24 volt vehicle electrical system circuitry, especially the primary and secondary portions of ignition systems. The test set shall be capable of withstanding secondary ignition system voltages of 45 KV, both polarities.

3.2.5 Tachometer overload. The low range of the tachometer circuit shall be capable of withstanding, without damage, an rpm input of twice the full scale value for 30 seconds when connected to an M151 truck ignition system. The tachometer shall indicate accurately and freely in all modes over the full scale after being overloaded.

3.2.6 Test set components.

3.2.6.1 Test set component limit. No more than 30 electronic/electrical components shall be used in the design and manufacture of the test set. All of the components shall be mounted upon a single printed circuit board. The mode selector switches shall be mounted on the printed circuit board and shall be counted as electronic/electrical components. The meter movement, and the circuit board itself, shall be counted as components. The test set shall be designed to operate, as specified herein, as an integral unit without addition or replacement of any electrical/electronic components.

3.2.6.2 Test set circuit(s) design. The number of components in the electronic/electrical assemblies, and the method of assembly, shall be directly related to the integrity and longevity of the test set. Therefore, the number of components used in the circuit design shall be held to a minimum and shall be as specified in 3.2.6.1.

3.2.7 Safety. The test set shall comply with the general safety requirements promulgated under OSHA 2206 that are applicable to the test set itself. Exceptions and additional requirements for safety shall be as specified in the contract (see 6.2.1 and 6.5).

3.2.8 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.9 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.2.10 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the test set. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in FED-STD-376.

3.3 Performance characteristics.

3.3.1 Accuracy. The accuracy of the test set, including the tachometer and dwell meter, shall be within three percent of full scale value when measured by calibrated instruments (see 3.3.1.1) at a temperature of 48° ±5°F and 87° ±5°F. Accuracy shall be within five percent of full scale value when measured by

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calibrated instruments at a temperature of 18° ±5°F and 122° ±5°F. Tachometer and dwell meter accuracy shall be demonstrated over an engine speed range of 400 to 3,000 rpm.

3.3.1.1 Calibration instruments. All test sets shall be calibrated against measuring and test equipment having at least ten times the accuracy required for the test set. The calibration and accuracy of measuring and test equipment shall be traceable to the National Bureau of Standards with calibration performed within the prior six month period.

3.3.2 Operating voltage. The test set shall perform within the accuracy limits specified herein when operating between 9.6 and 15.5 volts DC on 12 volt systems, and between 19.4 and 28.5 volts DC on 24 volt systems.

3.3.3 Position influence (balance). Position influence shall not exceed three percent of full scale value. Position influence shall be considered to be the maximum displacement of the meter pointer caused solely by a change in the inclination of the meter from the normal operating position. Normal operating position of the meter shall be when the test set is positioned on the rests, on the bottom of the case, with the meter window in a vertical plane.

3.3.4 Vibration. The test set in the carrying case, with the cover secured by the fastener(s), shall be capable of withstanding, without damage, six hours of vibration two hours in each of the three mutually perpendicular axes. One of the axes shall be parallel to the meter indicator shaft. The vibration levels shall be at a single amplitude of .010 inches (double amplitude: 0.020 total excursion), with the frequency varying from 10 hertz to 60 hertz traversed once per minute. After test, the accuracy shall remain within the tolerances specified in 3.3.1.

3.3.5 Service life. The test set shall have a demonstratable service life of at least 100 uses per month for two years. Service life shall be demonstrated in an accelerated life test consisting of 40 hours of operation, with the connection between the test set and the ignition system interrupted once per minute on various ignition systems and using various functions.

3.4 Temperature and humidity conditions.

3.4.1 Storage. The low and high temperature storage effect on the test set (stored in the carrying case) shall not change the accuracy more than one percent of the reading obtained, as specified in 3.3.1. The test set shall be stored for 24 hours at not higher than -20°F, followed by return to ambient, with an accuracy check, followed by 24 hours at not lower than +130°F and then a return to ambient, with an accuracy check.

3.4.2 Humidity. The humidity effect on the test set (without the carrying case) shall not change the accuracy more than one percent of the correct reading obtained, as specified in 3.3.1. Humidity conditions shall be 85 to 90 percent relative humidity at an air temperature of 75° to 85°F, air movement negligible.

3.5 Components.

3.5.1 Meter. The test set shall be provided with one meter with a movement of at least 100 degrees. The meter shall have four scales with ranges and markings as follows:

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Low rpm scale - 0 to at least 1,500 rpm
minor graduations every 20 rpm,
numbered every 300 rpm.

High rpm scale - 0 to at least 6,000 rpm
minor graduations every 100 rpm,
major graduations every 500 rpm,
numbered every 1,000 rpm.

Dwell scale - 0 to at least 90 degrees
minor graduations every two degrees,
numbered every 10 degrees.

Point resistance - good/bad scale based on the voltage drop across the closed points as follows:

	Bad Range	Good Range
Eight cylinder	resistance based on more than 0.6 V drop	less than 0.5 V drop
Six cylinder	resistance based on more than 0.35 to 0.40 V drop	less than 0.35 V drop
Four cylinder	resistance based on more than 0.2 to 0.3 V drop	less than 0.2 V drop

The meter face shall be protected with a clear plastic lens. The edges of the lens shall be sealed to prevent the entrance of dirt, dust, and moisture. Both sides of the lens shall be coated with anti-static solution to prevent static charges from influencing meter movement. A screw type zero adjustment for the meter shall be provided on the front of the test set housing.

3.5.2 Function switches. Function selector switches or knobs shall be provided on the front of the test set housing to select: low and high rpm, dwell, point resistance, and cylinder selector switch for four, six, and eight cylinder engines. The test set shall be designed to test 12 and 24 volt ignition systems without operator or manual selection or switching. The automatic switching for 12 and 24 volts primary ignition current shall be performed by solid state circuits within the test set.

3.5.3 Housing. The test set components shall be contained in a housing fabricated from metal or ABS plastic, or a combination thereof. The housing shall be sealed to protect the electrical and electronic components, operating mechanisms, meter movement, meter panel, and meter face from dust and moisture. A storage compartment shall be provided in the test set housing for storage of the test leads, connectors, and two commercial operator's handbooks. The handbooks and other printed material may also be stored in the carrying case (see 3.6 and 3.7). The bottom of the housing shall be provided with four oil resistant rubber or neoprene rests.

3.5.4 Test leads. One duplex stranded wire lead shall be provided. No additional leads, directly and permanently connected to the test set, shall be permitted. This restriction does not apply to distribution connections. The

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duplex lead shall remain joined from the test set case outward to the point where separation of the power/ignition system lead and the ground lead is necessary to accommodate test lead hook-up. The test lead shall be at least four feet long, with spring clip connectors and color coded boots. Primary ignition system positive (+) shall be red booted and ground negative (-) shall be black booted. The test lead connection to the test set shall be protected against damage with a strain relief grommet where the duplex lead passes through the test set housing.

3.6 Carrying case. A metal or high impact plastic carrying case shall be provided to carry the test set and provide a degree of protection against damage during transit in repair vehicle storage compartments. The case shall have a carrying handle which folds flat against the case. The case may also be used to store two commercial operator's handbooks. At least one fastener shall be provided to secure the lid in the closed position when the case and test set is being carried.

3.7 Commercial operator's handbook. Two copies of a commercial operator's handbook shall be furnished with each test set. The handbook shall contain complete instructions for hook-up to each of the ignition systems specified in 3.2, including military shielded systems. The handbook shall include complete operational and hook-up instructions as well as calibration instructions and procedures.

3.8 Marking. The test set housing exterior shall be marked with the following information as specified in MIL-STD-130:

Test Set, Tachometer/Dwell
NSN 4910-00-788-8549
Manufacturer:
Contract No.:
Date of Manufacture:

3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.10 Workmanship. Standards of workmanship shall assure that the test set shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, first article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (d) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Accuracy test (see 4.6.2).
(c) Accelerated life test (see 4.6.11).
(d) Packaging inspection (see 4.7).

4.4.1 Sampling. For sampling purposes, lot size shall be limited to no more than 150 test sets or one month's production, whichever is less. With the exception of the special sampling requirements of 4.4.2, sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination	II	1.0
Accuracy test	II	1.0
Packaging inspection	S-4	4.0

4.4.2 Accelerated life test sampling. The accelerated life test of 4.6.11 shall be performed on one test set selected at random from each lot of 150 sets or each month's production, whichever is greater. Failure of a test set to meet the accelerated life test requirement shall be cause for deferring further submission of test sets for Government acceptance until the contractor has completed the following actions, and three successive randomly selected test sets from the represented lot or month's production have completed the accelerated life test without failure. The contractor shall:

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(a) Conduct a failure analysis study, performing a dimensional, physical, and visual examination, as required, of the component(s) or condition(s) suspected to be the cause of failure.
(b) Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect in future production.
(c) Examine test sets awaiting submittal for Government acceptance, partially assembled test sets, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect(s) is purged from the inventory and not presented to the Government for acceptance.
(d) Provide objective evidence of the results of the failure analysis and corrective actions to the Government representative prior to submitting the reconditioned test sets for retest.

4.5 Product Examination. Visually and manually examine the test set to determine conformance with the requirements of 3.2, 3.2.6, and 3.5 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the test set has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection. A certificate of conformance may be accepted in lieu of inspection for the anti-static compound requirement of 3.5.1.

4.6 Tests.

4.6.1 Test stand. The test set shall be connected to an ignition system test stand. The test stand shall provide mountings for the various components of the ignition system types specified, listed in 3.2.1 through 3.2.3, shall provide for the proper interconnection of the components, and shall provide means for rotating the distributor to simulate the regulated engine speeds specified in 4.6.2. Power for the ignition system shall be provided by an appropriate automotive battery(s) augmented by a battery charger. The test stand shall include provisions for inducing secondary voltages to at least 45 KV. The voltages required above may be attained through selection and replacement of ignition system components, variation of spark gap, or any other suitable means which simulates the ignition system in the vehicle.

4.6.2 Accuracy test. Using the zero adjustment (see 3.5.1), the pointer shall be set at zero. The test set shall then be connected to an ignition system test stand, as specified in 4.6.1, and tested under the conditions specified below in the tachometer mode for breakerless systems, and in the tachometer, cam dwell, and point resistance modes for breaker point systems to verify conformance to 3.3.1, 3.3.2, and 3.5.1. Except for the performance test (see 4.6.9), the accuracy of the test set shall be verified on the M151A1 or M151A2 truck ignition system only. The dwell circuit and meter accuracy shall be measured by any method having an accuracy of within 0.3 degree. The point resistance circuit and meter accuracy shall be measured by any method having an accuracy of within 0.03 VDC. The tachometer circuit and meter accuracy shall be measured by any method having an accuracy of within 0.3 percent of the tachometer full scale value.

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(a) Engine speeds: 400, 750, 1,500, and 3,000 rpm.
(b) Primary coil voltages: 9.6 VDC and 15.5 VDC (for 12 volt systems). 19.4 VDC and 28.5 VDC (for 24 volt systems).

4.6.3 Operating temperature tests.

4.6.3.1 Low ambient temperature test. The test set, after being thermally stabilized at a temperature of +20°F (-0°, +5°F) and while at that temperature, shall be tested for accuracy in accordance with 4.6.2 to determine compliance with 3.3.1 and 3.3.2.

4.6.3.2 High ambient temperature test. After being thermally stabilized at a temperature of +120°F (-5°, +0°F), and while at that temperature, the test set shall be tested in accordance with 4.6.2 to determine compliance with 3.3.1 and 3.3.2.

4.6.4 Position influence test. Using the zero adjustment, the pointer shall be set at zero with the meter in the normal horizontal operating position. The test set shall then be turned to rest on the left side, the right side, and the top. The deviation of the pointer shall be noted at each position to determine compliance with 3.3.3.

4.6.5 Tachometer overload test. Using an M151A1 or M151A2 truck ignition system and the test stand, the low range of the tachometer circuit shall be subjected to an rpm input of twice the full scale value for a period of 30 seconds. Upon completion of this test, the tachometer shall be inspected for damage and tested in accordance with 4.6.2 to determine compliance with the requirements of 3.2.5. The tachometer shall indicate freely and accurately in all modes on all parts of the scale.

4.6.6 Electrical load test. The test set shall be connected through the test leads to each of the following electrical loads for five minutes:

(a) An electrical circuit carrying 120 VAC with a capacity of 30 amps.
(b) 24 VDC (normal polarity) provided by two fully charged 12 Volt, 500 amp-hour automotive batteries connected in series.
(c) 24 VDC (reverse polarity) provided by two fully charged 12 Volt, 500 amp-hour automotive batteries connected in series.
(d) Secondary ignition system voltage of 45 KV (normal polarity).
(e) Secondary ignition system voltage of 45 KV (reverse polarity).

The voltages for (d) and (e) above shall be provided by any of the high energy ignition systems listed in 3.2.2 capable of producing the required voltage in conjunction with the test stand. Upon completion of the test, the test set shall be inspected for damage and tested for accuracy in accordance with 4.6.2.

4.6.7 Vibration test. The complete test set in its carrying case, with the cover secured by the fastener(s), shall be subjected to vibration (simple harmonic motion) with an amplitude of 0.010 inches (0.020 inches total excursion) through a frequency range of 10 to 60 Hz, traversed once per minute. The test set shall be vibrated for two hours in each of the three mutually perpendicular axes, one of which shall be parallel to the meter indicator shaft, to determine compliance with 3.3.4. Upon completion of the test, the test set shall be inspected for damage and tested for accuracy in accordance with 4.6.2.

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4.6.8 Storage tests.

4.6.8.1 Low temperature storage test. The test set, in its carrying case, shall be thermally stabilized at a temperature of -20°F, or less, and kept at that temperature for a period of 24 hours. At the end of this time, the test set shall be allowed to stabilize at a temperature of +50°F to +85°F. The test set shall then be inspected for damage and tested for accuracy in accordance with 4.6.2 (see 3.4.1).

4.6.8.2 High temperature storage test. The test set, in its carrying case, shall be thermally stabilized at a temperature of +130°F, or more, and kept at that temperature for a period of 24 hours. At the end of this time, the test set shall be allowed to stabilize at a temperature of +50°F to +85°F. The test set shall then be inspected for damage and tested for accuracy in accordance with 4.6.2 (see 3.4.1).

4.6.8.3 Humidity test. The test set shall be subjected to 24 hours of exposure at a relative humidity of 85 to 90 percent in an air temperature of +75° to +80°F, air movement negligible. While in this condition, the test set shall be tested for accuracy, in accordance with 4.6.2, to determine compliance with 3.4.2.

4.6.9 Performance test. The test set shall be tested for function and accuracy in accordance with 4.6.2 using each of the ignition system types specified in 3.2.1 through 3.2.3. The test set shall be operated at each voltage and speed specified, and shall show no evidence of instability or loss of function during the test.

4.6.10 Military vehicle test. The test set, after completion of the test in 4.6.9, shall be connected to the ignition system on either a Truck, Utility, 1/4 ton, M151A1 (NSN 2320-00-763-1092) or M151A2 (NSN 2320-00-177-9258), and either a Truck, Cargo, 1-1/4 Ton, 4 X 4, M880 (NSN 2320-00-579-8942) or 4 X 2, M890 (NSN 2320-00-579-8991). The test stand may not be used for this test. Engine speed and dwell (for the M151A1 or M151A2) shall be measured at speeds in the low, mid, and high portions of the scale when the test set is at its low rpm setting. The test shall be repeated with polarity reversed. Any evidence of instability, loss of function, or damage to either the vehicle or the test set, attributable to the test set, shall constitute failure of this test.

4.6.11 Accelerated life test. Using the engine ignition system test stand, the test set shall be subjected to a 40 hour accelerated life test. Throughout the test, the electrical connection between the test set and the ignition system on the test stand shall be interrupted for 0.5 to 1.0 second once every minute for a total of 2,400 cycles. The test shall be divided as follows:

(a) Fifteen hours on any of the ignition system types specified in 3.2.2. Primary coil input voltage shall be 15.5 VDC, the distributor shall be an 8 lobe type, and the simulated engine speed shall be 1,500 rpm. This portion of the test shall be for the tachometer function only.

(b) Fifteen hours on any of the ignition system types specified in 3.2.1. Primary coil input voltage shall be 15.5 VDC, the distributor shall be a 6 lobe type, and the simulated engine speed shall be 1,500 rpm. This portion of the test shall be composed of ten hours

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of point dwell measurement and five hours of point resistance measurement.

(c) Ten hours on a M151A1 or M151A2 truck ignition system. Primary coil input voltage shall be 28.5 VDC, and the simulated engine speed shall be 1,500 rpm. This portion of the test shall be comprised of five hours of tachometer operation and five hours of dwell measurement.

The test set shall complete the full 40 hours of testing without evidence of instability, damage, or loss of function. The test set shall then be tested for accuracy in accordance with 4.6.2.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation, Levels A or B packing, and military marking shall be in accordance with MIL-E-17555.

5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The test set covered by this specification consists of a dwell meter circuit and a tachometer. The test set derives its total operating power from the vehicle electrical system. The dwell meter circuit of the unit is intended to be used for measuring distributor cam dwell and for the adjustment of breaker points on internal combustion engines. Indications of worn distributor shaft bushings, bouncing of breaker points, resistance between breaker points, or in the wiring and connections of the distributor primary circuit are also functions of the instrument. The tachometer is used for measuring engine rpm, carburetor idle settings, setting the engine governor, and for other uses where it is desired to know engine speed.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Safety requirements if different (see 3.2.7).
(d) Painting and finishing if different (see 3.9).
(e) First article inspection if different (see 4.3).
(f) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

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6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items should be identified as shown in MIL-M-7298.

6.4 Military engine adapter set. An adapter set, identified as Adapter Set, Engine Electrical Test (NSN 4910-00-348-7600) and covered by MIL-A-13092, may be used to interface the test set covered by this specification with the 24 volt M151A1 and M151A2 ignition system (see 3.2). The military engine adapter set consists of the following parts:

Part	Army Drawing No.
Adapter, Primary Circuit	B7540877
Adapter, Spark Plug	B7540884
Adapter, Ignition Unit	B7540895
Adapter, Testing, Regulator	B7541200
Adapter, Testing, Generator	C7541209

The above military engine adapter set is not supplied as a part of the test set covered by this specification. The above identification is furnished as a matter of information only.

6.5 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Navy - YD
Air Force - 99

Preparing activity:

Army - AL

Project No. 4910-0609

Review activity:

Air Force - 82

User activity:

Army - AT

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	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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